Alpha-Ramsey Theory

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Overview

1. Nonstandard Analysis
2. The Alpha-Theory
3. Alpha-Trees
4. Alpha-Ramsey Theory
5. Applications

Nonstandard analysis

(1961) Nonstandard analysis was introduced by Robinson to reintroduced infinitesimals and infinite numbers into analysis.

(1961) Using model theory, Robinson gave a rigorous development of the calculus of infinitesimals.

- Ultrafilters
- Ultrapowers
- $*$-transfer principle

(2003) Benci and Di Nasso in [2] have introduced a simplified presentation of nonstandard analysis called the Alpha-Theory.

The Alpha-Theory

Idea

Extend the natural numbers $\mathbb{N}$ by adjoining a new number α that behaves like a very large natural number.

(1958) Laugwitz [3] adjoin a new symbol Ω and assume that a 'formula' is true at Ω if it is true for all sufficiently large natural numbers.

(2003) Alpha-Theory approach can be seen as a strengthening of the Ω-Theory.

(2003) Alpha-Theory conservatively extends *ZFC* with five new axioms which describe a new symbol α.

Sequences

Definition

A **sequence** is any function f whose domain is $\mathbb{N}$.

Example

$$(.9, .99, .999, .9999, \dots)$$

$$(1, 1/2, 1/3, 1/4, 1/5, \dots)$$

$$(\mathbb{R}, \mathbb{R}^2, \mathbb{R}^3, \mathbb{R}^4, \mathbb{R}^5, \dots)$$

$$(\alpha, \alpha, \alpha, \alpha, \alpha, \alpha, \dots)$$

Axiom 1, 2 & 3

Axiom (Extension)

For all sequences $f : \mathbb{N} \to X$ there a unique element $f[\alpha]$, called the "ideal value of f."

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Axiom (Pair)

The ideal value of $(\{x_0, y_0\}, \{x_1, y_1\}, \{x_2, y_2\}, \{x_3, y_3\}, \dots)$ is

$$\{x[\alpha], y[\alpha]\}.$$

Axiom 4

Axiom (Composition)

Let $(x_1, x_2, x_3, \dots)$ and $(y_1, y_2, y_3, \dots)$ be sequences and f be a function such that

$$(f(x_1), f(x_2), f(x_3), \dots) \text{ and } (f(y_1), f(y_2), f(y_3), \dots)$$

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If $x[\alpha] = y[\alpha]$ then

$$(f(x_1), f(x_2), f(x_3), \dots) \text{ and } (f(y_1), f(y_2), f(y_3), \dots)$$

have the same ideal values.

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Definition

The elements of ${}^*\mathbb{N} \setminus \mathbb{N}$ are called **nonstandard hypernatural numbers**.

The $*$-Transform

Theorem (Proposition 2.2, [2])

For all sets A and B the following hold:

1. $A = B \iff {}^*A = {}^*B$
2. $A \in B \iff {}^*A \in {}^*B$
3. $A \subseteq B \iff {}^*A \subseteq {}^*B$
4. ${}^*\{A, B\} = \{{}^*A, {}^*B\}$
5. ${}^*(A, B) = ({}^*A, {}^*B)$
6. ${}^*(A \cup B) = {}^*A \cup {}^*B$
7. ${}^*(A \cap B) = {}^*A \cap {}^*B$
8. ${}^*(A \setminus B) = {}^*A \setminus {}^*B$
9. ${}^*(A \times B) = {}^*A \times {}^*B$

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Recall that in ZFC a binary relation R between two sets A and B is identified with the set

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Theorem (Proposition 2.3, [2])

If $f : A \to B$ is a function then $^*f : {}^*A \to {}^*B$ is also a function.

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Example

$$\{0, 3, 5\} \sqsubseteq \{0, 3, 5, 8, 9, 12\}$$

$$\{1, 15\} \sqsubseteq \{1, 15, 25, \dots\}$$

$$\{1, 15\} \not\sqsubseteq \{0, 3, 5, 8, 9, 12\}$$

$$\{1, 15\} \not\sqsubseteq \mathbb{N}$$

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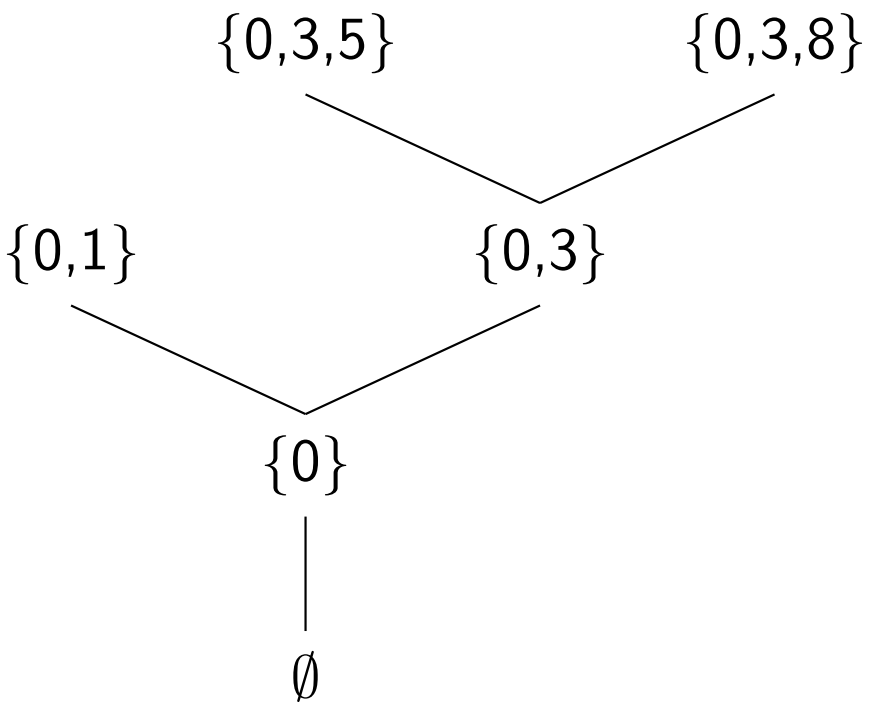

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The **stem of** T, if it exists, is the first place where the tree branches. If T has a stem we denote it by $st(T)$.

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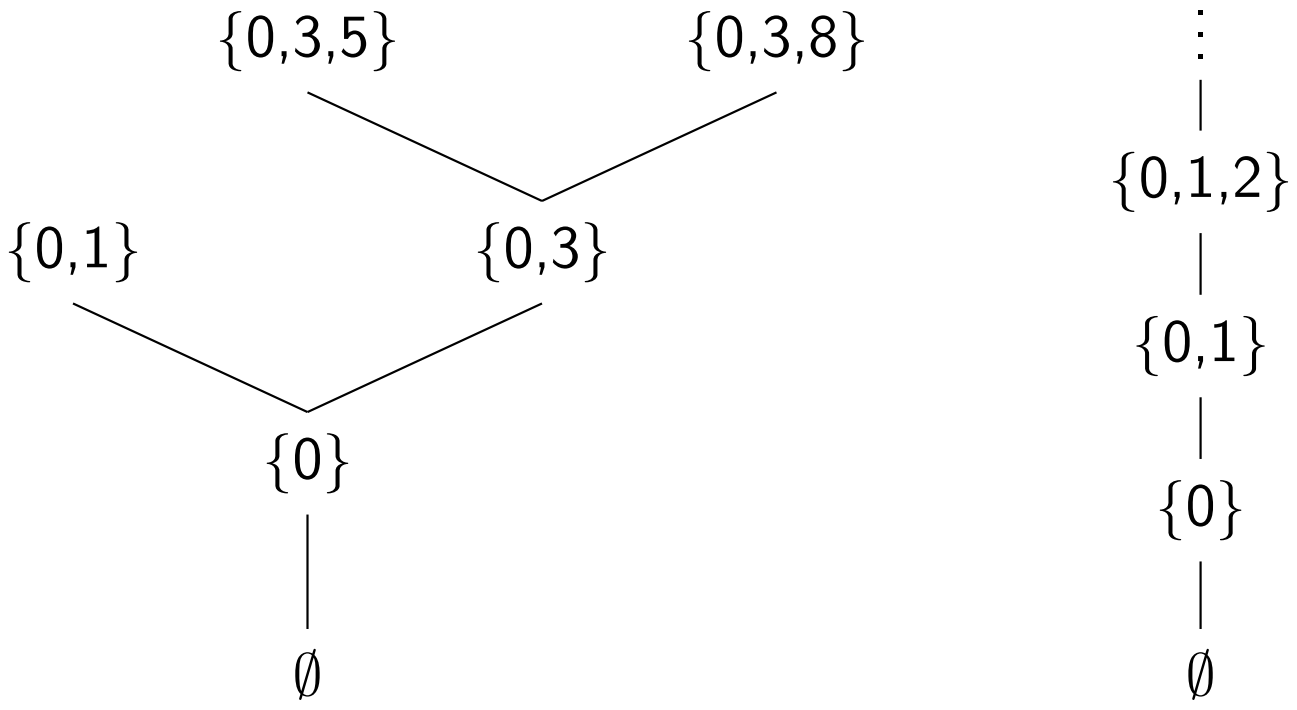

Alpha Trees

Notation

For $s \in T$, we use the following notation

$$T/st(T) = \{t \in T : st(T) \sqsubseteq t\}.$$

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Example

The tree

$$[\mathbb{N}]^{<\infty} = \{s \subseteq \mathbb{N} : |s| < \infty\}$$

is an α-tree.

Alpha-Ellentuck topology

Theorem (T. [1])

If S and T are α-trees and $S \cap T \neq \emptyset$ then $S \cap T$ is an α-tree.

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For a tree T we let $[T] = \{X \in [\mathbb{N}]^\infty : \forall s \sqsubseteq X, s \in T\}$.

Definition

The α-**Ellentuck topology on** $[\mathbb{N}]^\infty$ is the topology generated by

$$\{[T] : T \text{ is an } \alpha\text{-tree}\}.$$

Alpha-Ramsey

Definition

$\mathcal{X} \subseteq [\mathbb{N}]^\infty$ is α-**Ramsey** and for all α-trees T there exists an α-tree $S \subseteq T$ with $st(S) = st(T)$ such that one of the following holds:

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The Alpha-Ellentuck theorem

Definition

$\mathcal{X} \subseteq [\mathbb{N}]^\infty$ has **the Baire property with respect to the α-Ellentuck topology** if there exists a meager set $\mathcal{M}$ and and open set $\mathcal{O}$ with respect to the α-Ellentuck topology such that

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Theorem (T. [1], α-Ellentuck theorem)

$\mathcal{X}$ is α-Ramsey if and only if $\mathcal{X}$ has the Baire property with respect to the α-Ellentuck topology.

Applications

1. New characterization of Ramsey sets (T. [1], 2016)
2. Ramsey Theorem (1929)
3. Nash-Williams Theorem (1965)
4. Silver Theorem (1970)
5. Galvin-Prikry Theorem (1973)
6. Local Ramsey Theory (1972, Louveau)
7. Ultra-Ramsey Theory (2010, Todorcevic)
8. Abstract ultra-Ramsey Theory (T. [1], 2016)
9. Abstract local Ramsey Theory (T. [1], 2016)
10. New characterization of the strong Cauchy infinitesimal principle. (T. [1], 2016)

Thank you for your attention!

- Trujillo, *From abstract Alpha-Ramsey theory to abstract ultra-Ramsey theory.* 2016.
- Benci & DiNasso, *Alpha-Theory*, Expositiones Mathematicae, 2003.
- Laugwitz, Ω-*Theory* 1959.